TASK 1 REPORT — Rating Prediction via Prompting

1. Introduction

The objective of Task 1 was to design and evaluate different prompting strategies for predicting Yelp review ratings (1–5 stars) using an LLM. The goal was to study the effect of prompt engineering on accuracy, JSON reliability, and consistency.

2. Dataset

The Yelp Reviews Dataset from Kaggle was used. A subset of 200 reviews was sampled containing only 'text' and 'stars' columns.

3. Experimental Setup

Model: deepseek/deepseek-chat (OpenRouter API)

Temperature: 0

A regex-based JSON extraction method ensured 100% valid JSON outputs.

4. Prompting Approaches

1. Zero-Shot Prompting: No examples provided.

2. Few-Shot Prompting: Three labeled examples included.

3. Chain-of-Thought Prompting: Internal reasoning, JSON-only output.

5. Evaluation Metrics

- Accuracy (exact match)

- JSON Validity Rate

- Consistency (manual comparison)

6. Results

Zero-Shot: Accuracy 0.655, JSON Validity 1.000

Few-Shot: Accuracy 0.665, JSON Validity 1.000

Chain-of-Thought: Accuracy 0.635, JSON Validity 1.000

7. Analysis

Few-shot prompting performed best. Zero-shot was close. Chain-of-thought sometimes over-analyzed reviews, reducing accuracy slightly. Strict JSON rules ensured perfect validity.

8. Conclusion

Prompt engineering significantly impacts performance. Few-shot prompting offers the best accuracy and consistency for this task.